

BIOS 755: Generalized Estimating Equations (GEEs) or Marginal Models for Longitudinal Data

Alexander McLain

March 3, 2026

Longitudinal model for non-normal data

- ▶ Longitudinal models for normal data are heavily influenced by the multi-variate normal (MVN) distribution
- ▶ In fact, the MVN distribution makes most of what we did in our linear models section possible.
- ▶ The MVN distribution allows us to relate multiple variables through their covariances.
- ▶ With non-normal data, this isn't as easy.
 - ▶ We would have to assume higher-order relationships, i.e., How does $P(Y_{i2} = 1 | Y_{i1} = 1)$ vary by Y_{i3}
- ▶ What are we going to do?

Full distributions with non-normal data

- ▶ Unfortunately, with discrete response data, there is no analog of the multivariate normal distribution.
- ▶ In the absence of a 'convenient' likelihood function for discrete data, there is no unified likelihood-based approach for marginal models.
- ▶ Recall: In linear models for normal responses, specifying the means and the covariance matrix fully determines the distribution of the data and the likelihood.
- ▶ This is not the case with discrete response data.

Marginal Models

- ▶ One approach is to specify the marginal distribution at each time point:

$$Y_{ij} \text{ for } j = 1, \dots, n_i$$

along with some assumptions about the covariance structure of the observations.

- ▶ Marginal models avoid some distributional assumptions used with other methods (e.g., mixed models).
- ▶ They don't make a complete assumption on the full distribution, why they are "marginal".
- ▶ Marginal models are conditional on the covariates and the covariance structure only (i.e., no random effects needed).

Marginal Models

- ▶ The basic premise of marginal models is to make inferences about population averages.
 - ▶ What is happening to the average? vs What is happening to each subject?
- ▶ Marginal models will look at the impact of exposures on group A vs group B, instead of the impact of an exposure of a subject changing from group A to group B.
- ▶ For linear models, all coefficients had the same interpretation; for GLM, this is no longer the case (we'll discuss this later).
- ▶ Marginal models are primarily used to make inferences on the impact covariates have on the population.

Assumptions of Marginal Models

- ▶ With marginal models we make the following assumptions:
 1. The **marginal expectation** of the response, $E(Y_{ij}) = \mu_{ij}$, depends on explanatory variables, X_{ij} , through a known link function

$$\eta_{ij} = g(\mu_{ij}) = \mathbf{X}_{ij}\beta$$

2. The **marginal variance** of Y_{ij} depends on the marginal mean according to

$$\text{Var}(Y_{ij}) = v(\mu_{ij})\phi$$

where $v(\mu_{ij})$ is a known 'variance function' and ϕ is a scale parameter that may need to be estimated. (You'll have limited impact on this portion for most models)

3. The covariance between Y_{ij} and Y_{ik} is a function of the means and additional **correlation parameters** that will also need to be estimated. (Similar to covariance pattern models.)

Examples of Marginal Models

Continuous responses:

1. $\mu_{ij} = \eta_{ij} = \mathbf{X}_{ij}\beta$, i.e., linear regression
2. $\text{Var}(Y_{ij}) = \phi$, i.e., homogeneous variance.
3. $\text{Corr}(Y_{ij}, Y_{ik}) = \alpha^{|k-j|}$ ($0 \leq \alpha \leq 1$), i.e., autoregressive correlation.

Examples of Marginal Models

Binary responses:

1. $\text{logit}(\mu_{ij}) = \eta_{ij} = \mathbf{X}_{ij}\boldsymbol{\beta}$, i.e., logistic regression
2. $\text{Var}(Y_{ij}) = \mu_{ij}(1 - \mu_{ij})$, i.e., Bernoulli variance.
3. $\text{Corr}(Y_{ij}, Y_{ik}) = \alpha_{jk}$ ($0 \leq \alpha_{jk} \leq 1$), i.e., unstructured correlation.

Examples of Marginal Models

Count responses:

1. $\log(\mu_{ij}) = \eta_{ij} = \mathbf{X}_{ij}\boldsymbol{\beta}$, i.e., Poisson regression
2. $\text{Var}(Y_{ij}) = \mu_{ij}\phi$, i.e., extra-Poisson variance.
3. $\text{Corr}(Y_{ij}, Y_{ik}) = \alpha$ ($0 \leq \alpha \leq 1$), i.e., compound symmetry correlation.

Similarities with GLMs

- ▶ The assumptions of marginal models are similar to Generalized Linear Models (GLMs).
 - ▶ both have a systematic component
 - ▶ both have a formula
 - ▶ the variance of both is **usually** specified by a distribution (i.e., Bernoulli, Poisson, etc.)
- ▶ Marginal models add a covariance structure to the specification.
 - ▶ They're similar to a GLM with covariance.

Similarities with GLMs

- ▶ The assumptions of marginal models are similar to Generalized Linear Models (GLMs).
 - ▶ both have a systematic component
 - ▶ both have a formula
 - ▶ the variance of both is **usually** specified by a distribution (i.e., Bernoulli, Poisson, etc.)
- ▶ Marginal models add a covariance structure to the specification.
 - ▶ They're similar to a GLM with covariance.
- ▶ Marginal models don't specify a distribution, except through the relationship between the mean and the variance. (not likelihood-based)
- ▶ Marginal models don't specify the entire joint distribution of the data.

Interpretations in a Marginal World

- ▶ The regression parameters β have 'population-averaged' interpretations:
 - ▶ describes the effect of covariates on the average responses
 - ▶ think of them as comparing the means in sub-populations
- ▶ In linear regression, what happens between the groups is the same as what would happen to an individual going from one group to the other. Here, that's not the case.
- ▶ The increase in the probability of a heart attack between 40-year-olds and 50-year-olds is not the same as an individual increase in the probability of a heart attack when aging from 40 to 50.

Generalized Estimating Equations

- ▶ Since there is no 'convenient' or natural specification of the joint multivariate distribution of $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, \dots, Y_{in})$ for marginal models when the responses are non-normal, we use an alternative to maximum likelihood (ML) estimation.
- ▶ Liang and Zeger (1986) and Zeger, Liang, and Albert (1988) proposed such a method based on the concept of 'estimating equations.' This work comes from:
 - ▶ Wedderburn (1974) Quasi-likelihood functions, generalized linear models, and the Gauss-Newton method, *Biometrika*, 61: 439–447
 - ▶ McCullagh (1983) Quasi-Likelihood Functions, *The Annals of Statistics*, 11: 59–67
- ▶ This provides a general and unified approach for analyzing discrete and continuous responses with marginal models.

Generalized Estimating Equations

- ▶ **Generalized Estimating Equations (GEEs)** are a statistical technique for analyzing correlated or clustered data.
- ▶ GEEs extend GLMs to correlated data by introducing a working correlation structure that accounts for the relationship between observations within a cluster.
- ▶ GEEs use quasi-likelihood methods rather than full likelihood methods to estimate the model parameters.
- ▶ **Working Correlation Structure:** This is a key component of the GEE approach. The working correlation structure is a mathematical representation of how data points within a cluster are related.
 - ▶ Common structures include independence, exchangeable, and autoregressive.

Fitting Marginal Models

- ▶ Let $\mathbf{Y}_i = (Y_{i1}, Y_{i2}, \dots, Y_{in})$ be a vector of correlated responses for the i th subject ($i = 1, \dots, N$).
- ▶ Then, an estimate of β can be obtained as the solution to the following 'generalized estimating equation'

$$\sum_{i=1}^n \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{Y}_i - \mu_i) = \mathbf{0} \quad (1)$$

where $\mathbf{D}_i = \partial \mu_i / \partial \beta$

- ▶ $\mathbf{V}_i$ is a 'working' covariance matrix, i.e. $\mathbf{V}_i \approx \text{Cov}(\mathbf{Y}_i)$, which **is a function of ϕ and α** .

Fitting Marginal Models

- ▶ Generalized estimating equations depend on β , ϕ (the variance parameter(s)), and α (the correlation parameter(s)).
- ▶ Because the GEEs depend on both mean and covariance parameters, an iterative two-stage estimation procedure is required:
 1. Given current estimates of (α, ϕ) , an estimate of β is obtained as the solution to (1) on the previous slide.
 2. Given current estimate of β estimates of α and ϕ are obtained based on the residuals,

$$r_{ij} = Y_{ij} - \hat{\mu}_{ij}$$

Properties of GEE estimators

Assuming α and ϕ are consistent:

- ▶ $\hat{\beta}$ is a consistent estimate of β (with high probability $\hat{\beta}$ is close to β for large n).
- ▶ In large sample, $\hat{\beta}$ has a multivariate normal distribution.
- ▶ $\text{Cov}(\beta) = \mathbf{F}^{-1} \mathbf{G} \mathbf{F}^{-1}$ where

$$\begin{aligned} \mathbf{F} &= \sum_{i=1}^n \mathbf{D}_i^{-1} \mathbf{V}_i \mathbf{D}_i^{-1} \\ \mathbf{G} &= \sum_{i=1}^n \mathbf{D}_i \mathbf{V}_i^{-1} \text{Cov}(\mathbf{Y}_i) \mathbf{V}_i^{-1} \mathbf{D}_i \end{aligned}$$

This is called the “empirical” or “sandwich” variance estimator.

Properties of GEE estimators

- ▶ $\hat{\beta}$ is consistent even if the covariance of $\mathbf{Y}_i$ has been misspecified (robust).
- ▶ The variance of $\hat{\beta}$ can be estimated by $\mathbf{F}^{-1}$ or $\mathbf{F}^{-1}\mathbf{GF}^{-1}$.
 - ▶ $\mathbf{F}^{-1}$ is the 'model-based' estimator.
 - ▶ $\mathbf{F}^{-1}\mathbf{GF}^{-1}$ is the 'empirical' or 'sandwich' estimator.
- ▶ The standard errors of $\hat{\beta}$, as measured by $\mathbf{F}^{-1}\mathbf{GF}^{-1}$ are asymptotically valid even when the correlation structure is incorrect.
 - ▶ Why model the correlation, then?

F versus G

Both model-based and sandwich-based estimators are useful in different situations:

- ▶ **Sandwich based** is best to use when
 - ▶ sample size is relatively large (several hundred subjects or more)
 - ▶ when the assumed model for the covariances is questionable.
- ▶ **Model based** is best to use when
 - ▶ sample size is smaller
 - ▶ small number of clusters.
- ▶ Model-based needs the correlation/covariance to be modeled correctly.

Practical Application Steps

- ▶ **Defining the Model:** Specify a model that includes independent variables (predictors) and a dependent variable (outcome), choosing a link function and distribution that match the nature of the data (e.g., binary, count, continuous).
- ▶ **Choosing a Working Correlation Structure:** Select an appropriate correlation structure (e.g., independent, exchangeable, autoregressive).
- ▶ **Estimation:** Use SAS/R to estimate the model's parameters using the quasi-likelihood approach, which does not require specifying the full distribution of the outcome.
- ▶ **Interpreting Results:** The focus is on interpreting population-averaged effects, with robust standard errors used to assess the statistical significance of predictors.
- ▶ **Sensitivity Analysis:** May perform sensitivity analyses with different correlation structures to check the robustness of the results.

GEE limitations

- ▶ Likelihood-based methods are NOT available for testing fit, comparing models, and conducting inferences about parameters.
- ▶ Sandwich-based estimators are more variable than parametric ones.
- ▶ Sandwich-based standard errors underestimate the true ones unless the sample size has several hundred subjects or more.
- ▶ More ideal for balanced data.
- ▶ Missing data needs to be handled a little more carefully.

GEE Advantages

- ▶ **Flexibility:** GEEs can handle many data types and link functions, making them highly versatile.
- ▶ **Robustness:** They provide robust estimates even when the correlation structure is misspecified, as long as the mean model is correctly specified.
- ▶ **Ease of Use:** Many statistical software packages support GEE analysis, facilitating its application in various research areas.
- ▶ **Causal inference:** Marginal Structural Models (MSM), which are an advanced method in causal inference with time-varying confounders, has connections with GEEs.